Verification and Validation Report: CEWS System

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1 Revision History

Date	Version	Notes
2023-03-27	0.1	Creation
2023-04-16	1.0	Completed Version 1.0

2 Symbols, Abbreviations and Acronyms

symbol	description
T	Test
CEWS	Cut Edge Weight Statistic
J_N	Normalized Multiclass Cut Edge Weight Measurement
D	Input Data Matrix
C	Input Class Matrix
FR	Functional Requirement
NFR	Non Functional Requirement

Contents

List of Tables

This document reviews the Verification and Validation specifications as outlined in the VnVPlan.

3 Functional Requirements Evaluation

- R1: Input
  - Description: CEWS System will read the input data.
  - Result: Pass
- R2: Calculation
  - Description: After reading the input data, CEWS System will perform the calculation and output the results.
  - Result: Unclear
  - Explanation: Issues with existing implementations may invalidate the test cases (see Section 6).
- R3: Reproducibility
  - Description: The CEWS System will generate the exact same output for the exact same dataset.
  - Result: Pass

4 Nonfunctional Requirements Evaluation

- NFR1: Accuracy
  - Description: The accuracy of the computed solutions should meet the level needed for data analysis.
  - Result: Unclear, due to testing results for R2 being unclear.
- NFR2: Portability
  - Description: The software should run any machine running the Windows 11 operating system.
  - Result: Pass

5 Test Results

This section reviews the results of the automated tests.

- T1
  - Result: Pass
- T2
  - Result: Pass
- T3
  - Result: Pass
- T4
  - Result: Pass
- T5
  - Result: Pass
- T6
  - Result: Pass
- T7
  - Result: Pass
- T8
  - Result: Pass
- T9
  - Result: Pass
- T10
  - See Section 6.1.
- T11
  - See Section 6.2.

- T12
  - Result: Pass
- T13
  - Result: Pass
- T14
  - Result: Pass
- T15
  - Result: Pass
- T16
  - Result:

Benchmark Dataset	Test Result
Breast Cancer	Pass
Glass Identification	Pass
Haberman	Pass
Image Segments	Pass
Ionosphere	Pass
Iris (Bezdek)	Pass
Iris Plants	Pass
Wine Recognition	Pass
Yeast	Pass

Table 1: T16 Results

6 Comparison to Existing Implementation

There are two existing implementations for different parts of the system. Note: the CEWS calculated with the graph from Cavanagh does not match the results from Zighed et al. (2005). It's possible there are errors in one or both of the implementations.

No accessible graph generation implementation has been able to reproduce the results in Zighed et al. (2005). The implementation used by the original authors does not run on modern operating systems. It is possible but unlikely that the benchmark datasets have changed since Zighed et al. (2005) was published.

Cavanagh implements the same algorithm used in Graph Module. However, the Cavanagh has not been independently verified. Also, Cavanagh has an error based on the SRS requirements as discussed in Section 7.2.

However, even if the graphs are not constructed using exactly the same method then it doesn't mean the program doesn't measure data complexity. In Zighed et al. (2005), the authors stated they achieved similar results with a Gabriel Graph or Minimum Spanning Tree. Another common data complexity measure, Fraction of Borderline Points uses a Minimum Spanning Tree and nearest neighbour graphs are often used as well. Lorena et al. (2020) As such the program may still be able to measure data complexity even if the program doesn't pass these tests.

6.1 T10

This test compares the program results to the results from Zighed et al. (2005).

Benchmark Dataset	Test Result
Breast Cancer	Pass
Glass Identification	Pass
Haberman	Pass
Image Segments	Pass
Ionosphere	Pass
Iris (Bezdek)	Pass
Iris Plants	Pass
Wine Recognition	Fail
Yeast	Pass

Table 2: T10 Results

The test fails on the Wine Recognition dataset with significant deviation (expected: 0.093, result: 0.268) which could indicate a legitimate error in the program.

6.2 T11

This test compares the program results to the results from Cavanagh.

Benchmark Dataset	Test Result
Breast Cancer	Pass
Glass Identification	Pass
Haberman	Pass
Image Segments	Pass
Ionosphere	Pass
Iris (Bezdek)	Fail
Iris Plants	Fail
Wine Recognition	Pass
Yeast	Fail

Table 3: T11 Results

7 Changes Due to Testing

7.1 Relative Neighbourhood Graph Algorithm

The Graph Module was converted to the simplest, naive algorithm for constructing a Relative Neighbourhood Graph in case the faster algorithms had slightly different behaviour in some cases and that was causing T11 and T16 to fail. Early implementations did pass T11 until changes made for Section 7.2.

7.2 Reproduciblility

During troubleshooting, the data was randomized to see if that was influencing the output of the existing implementations. There was a difference and there should not be for this application so the software requirements were revised to reflect that.

In a previous version, T16 would fail on the Yeast dataset but all T11 test cases would pass. Upon closer examination, it appears a numerical computation error (less than 1e-15) caused the ordering of the dataset to result in slightly different values for the distance. Both implementations had this error thus their result was the same. To fix the error in T16, an equality condition was converted to condition with a very small margin. This results in more edges being created in the relative neighbourhood graph. This fixed the error in T16 but then the values never longer match the existing implementation and T11 fails.

Notability, this case of distances between points being extremely similar exists only in some of the tested datasets. Only Yeast failed T16 originally but the changes also effected Iris Plants and Iris (Bezdek) as shown in T11. The other datasets appear to not have points that meet this condition, as they were unaffected by the change. Meeting the requirement R3 is more important for this implementation that matching existing implementations.

7.3 Duplicate Points

During testing, several deviations from expected results were traced back to the existence of duplicate points in the test data. Duplicate points have a distance of zero therefore always end up connected which stops the Graph Module from creating edges to non-duplicate close points. The program was change so points that are extremely close are considered to be duplicates and edges are not created between them.

7.4 Benchmark Dataset Processing

Some of the benchmark datasets from the UCI Machine Learning Repository had question marks in some fields to represent missing data. These benchmarks failed their tests due to the Input Module detecting them and generating an error. The samples with missing data were then removed from the datasets prior to running the program.

7.5 Traceability Between Test Cases and Requirements

	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
FR1	X	X	X	X	X	X	X	X	X							
FR2										X	X	X	X	X	X	
FR3																X
NFR1										X						
NFR2	X	X	X	X	X	X	X	X	X							

Table 4: Which test cases are supporting which requirements

8 Traceability Between Test Cases and Modules

	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
Control Module	X	X	X	X	X	X	X	X	X	X						X
Input Module	X	X	X	X	X	X	X	X	X	X						X
Output Module	X									X				X	X	X
Calculation Module	X									X						X
Graph Module	X									X	X	X	X			X

Table 5: Which test cases are supporting which modules

References

Tem Cavanagh. Relative neighborhood graph. https://github.com/temcavanagh/relativeNeighborhoodGraph. Accessed: 2023-04-02.

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